

Introduction

We hope to solve the equation

$$i\partial_t\psi = (-i\sigma_x\partial_x + m\sigma_z + f(x))\psi$$

where $\psi = (\psi_1\psi_2)^T$, and f is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor ψ , nearly all the same methods can be used to solve Dirac; we will use Strang splitting to approximate the solution by solving each term individually. Note that $\propto$ can usually be interpreted as "equal up to normalization."

Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize ψ_1 and ψ_2 into $|\psi_1\rangle$ and $|\psi_2\rangle$ as we know how from Schrödinger, and store the state ψ as

$$\begin{aligned} |\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k) |0\rangle + \psi_2(x_k) |1\rangle) \otimes |k\rangle \\ &= |0\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_1(x_k) |k\rangle \right) + |1\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_2(x_k) |k\rangle \right) \\ &\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle \end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for $|\psi_1\rangle$ and $|\psi_2\rangle$. Regarding measurement probabilities, if we measure only the register storing position, and leave the indicator qubit unmeasured, the probability of measuring $|k\rangle$ is $p(k) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$. This is justified as follows: the state $|\psi\rangle$ has

density matrix

$$\begin{aligned} |\psi\rangle\langle\psi| &\propto (|0\rangle\otimes|\psi_1\rangle + |1\rangle\otimes|\psi_2\rangle)(\langle 0|\otimes\langle\psi_1| + \langle 1|\otimes\langle\psi_2|) \\ &= |0\rangle\langle 0|\otimes|\psi_1\rangle\langle\psi_1| + |0\rangle\langle 1|\otimes|\psi_1\rangle\langle\psi_2| + |1\rangle\langle 0|\otimes|\psi_2\rangle\langle\psi_1| + |1\rangle\langle 1|\otimes|\psi_2\rangle\langle\psi_2| \end{aligned}$$

Measurement probabilities of the position register are given by the reduced density matrix obtained by tracing out the spinor qubit. Taking the partial trace, we get

$$\begin{aligned} \rho &\propto |\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2| \\ &= \sum_{l=0}^{N-1} \sum_{j=0}^{N-1} (\psi_1(x_l)\psi_1(x_j)^* + \psi_2(x_l)\psi_2(x_j)^*) |l\rangle\langle j| \end{aligned}$$

Thus the probability of measuring $|k\rangle$ at time t is $p(k) = \text{tr}(|k\rangle\langle k|\rho) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$ by orthonormality of the computational basis.

Transport Term

We concern ourselves with the equation $i\partial_t\psi = -i\partial_x\sigma_x\psi$. Recognizing that $\sigma_x = H\sigma_zH$, where, H is the Hadamard matrix, this system is equivalent to $\partial_t H\psi = -\partial_x\sigma_z H\psi$. Letting $\varphi := (\varphi_1, \varphi_2)^T = H\psi$, since σ_z is diagonal, the system is decoupled: $\partial_t\varphi = -\partial_x\sigma_z\varphi$. Equivalently,

$$\partial_t\varphi_1 = -\partial_x\varphi_1$$

$$\partial_t\varphi_2 = \partial_x\varphi_2$$

This is a first order linear transport equation, which has a known solution for initial conditions $\varphi_j^0(x) = \varphi_j(x, 0)$: $\varphi_1(x, t) = \varphi_1^0(x-t)$ and $\varphi_2(x, t) = \varphi_2^0(x+t)$. For the sake of implementation on a quantum computer, we solve using the

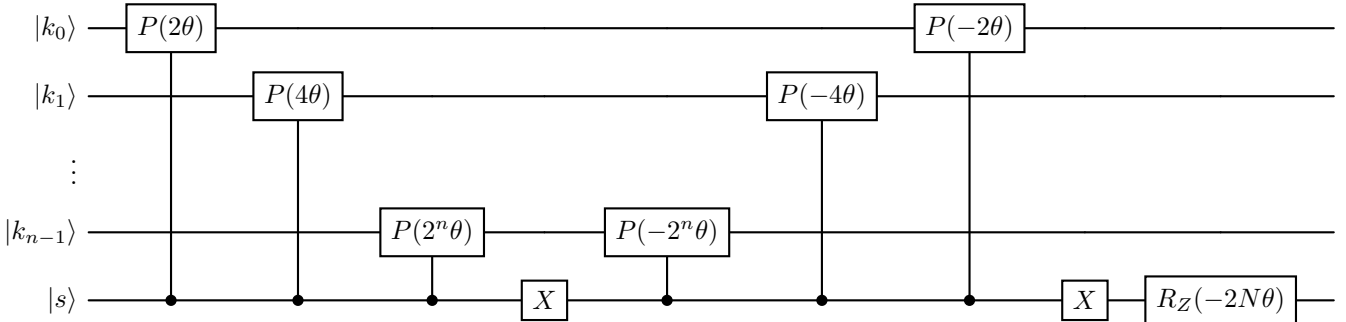
Fourier transform $\mathcal{F}$ to get

$$\varphi(x, t) = \begin{bmatrix} \mathcal{F}^{-1}(e^{-i\xi t} \mathcal{F}(\varphi_1^0)(\xi)) \\ \mathcal{F}^{-1}(e^{i\xi t} \mathcal{F}(\varphi_2^0)(\xi)) \end{bmatrix}$$

Hence on the quantum computer we map $|s\rangle \otimes |k\rangle \mapsto |s\rangle \otimes (QFT e^{(-1)^{s+1} i \xi_k \Delta t} QFT^\dagger |k\rangle)$, where as before, $\xi_k = \frac{\pi}{L}(2k - N)$. A new concern we have in Dirac is that any θ dependent global phase incurred in the application of $e^{i\theta k}$ is now a relative phase between φ_1 and φ_2 that we need to correct for. We write the phase shift operation more carefully with respect to global phase changes. Note that I've changed the implementation of $e^{ik} := \bigotimes_{j=0}^{n-1} P(2^j)$ to use phase gates, rather than differences of identity and Z gates. This makes global phase easier to keep track of.

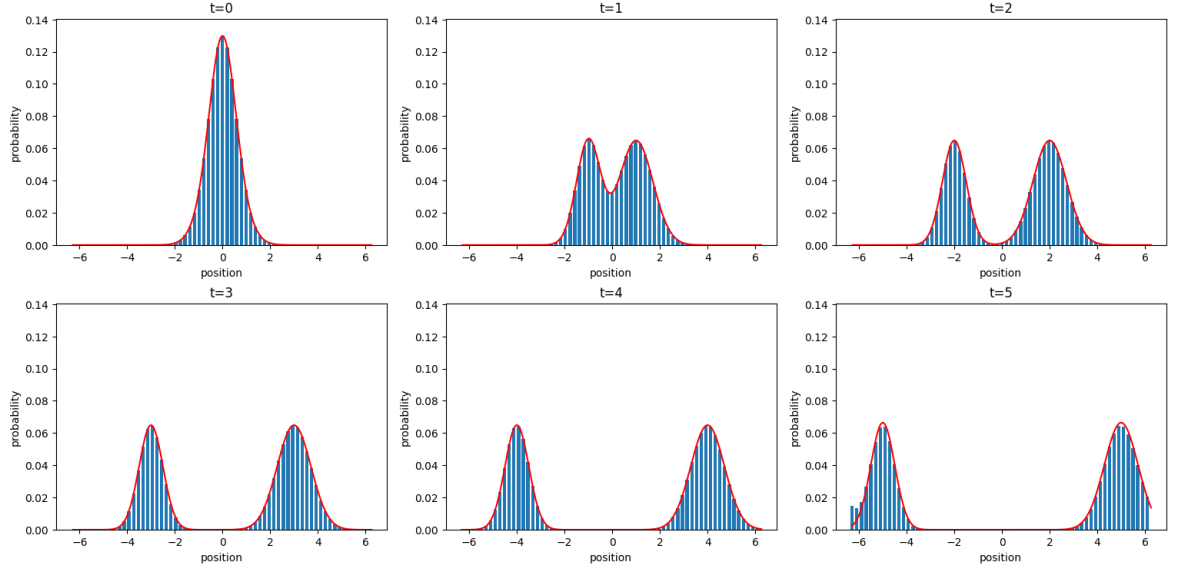
$$\begin{aligned} \exp\left((-1)^{s+1} i \frac{\pi}{L} (2k - N) \Delta t\right) &= \exp\left((-1)^s i N \frac{\pi}{L} \Delta t\right) \exp\left((-1)^{s+1} 2i \frac{\pi}{L} \Delta t k\right) \\ &= \exp\left((-1)^s i N \frac{\pi}{L} \Delta t\right) \bigotimes_{j=0}^{n-1} P\left((-1)^{s+1} 2^{j+1} \frac{\pi}{L} \Delta t\right) \end{aligned}$$

φ_1 picks up a phase of $e^{\frac{\pi i N \Delta t}{L}}$, and φ_2 picks up a phase of $e^{-\frac{\pi i N \Delta t}{L}}$, which we build in by applying $R_Z\left(\frac{-2\pi N \Delta t}{L}\right)$ to the spinor qubit. The quantum circuit we implement to apply the shift phase is below. $\theta = \frac{\pi}{L} \Delta t$.



Sandwiching this with QFT and $QFT^\dagger$ solves the linear transport equation,

which is certainly interesting on its own, so I have included a plot below of a solution with two Gaussians φ_1 and φ_2 .



As interesting as this is, our concern is with Dirac, and still we need to be able to map between $|\psi\rangle$ and $|\varphi\rangle$. This is fortunately very simple and reveals itself after a bit of algebra. Recalling that $\psi = H\varphi$ and $\varphi = H\psi$, and we have the state $|\varphi\rangle \propto |0\rangle|\varphi_1\rangle + |1\rangle|\varphi_2\rangle$

$$\begin{aligned}
 |\psi\rangle &\propto |0\rangle|\psi_1\rangle + |1\rangle|\psi_2\rangle \\
 &= \frac{|0\rangle}{\sqrt{2}}(|\varphi_1\rangle + |\varphi_2\rangle) + \frac{|1\rangle}{\sqrt{2}}(|\varphi_1\rangle - |\varphi_2\rangle) \\
 &= \frac{|0\rangle + |1\rangle}{\sqrt{2}}|\varphi_1\rangle + \frac{|0\rangle - |1\rangle}{\sqrt{2}}|\varphi_2\rangle \\
 &= (H \otimes I)|\varphi\rangle
 \end{aligned}$$

The same gate maps both directions, so we can transform between $|\varphi\rangle$ and $|\psi\rangle$ with a Hadamard gate on the spinor qubit. Thus our solution to $\partial_t \psi = -\sigma_x \partial_x \psi$ is given by simply applying H to the spinor qubit, following the work above for

the linear transport, then applying another H . This is the circuit returned by `transport()`.

Mass Term

We now concern ourselves with the mass term $i\partial_t\psi = m\sigma_z\psi$. Fortunately, σ_z is diagonal in the standard basis, so we have the following solution.

$$\psi(x, t) = \begin{bmatrix} e^{-imt}\psi_1^0(x) \\ e^{imt}\psi_2^0(x) \end{bmatrix}$$

On qubits we want to map $|0\rangle|\psi_1\rangle + |1\rangle|\psi_2\rangle \mapsto e^{-im\Delta t}|0\rangle|\psi_1\rangle + e^{im\Delta t}|1\rangle|\psi_2\rangle$.

This is implemented with a single $R_Z(2m\Delta t)$ on the spinor qubit.